Francisco Magno

Software Engineer · Agentic Engineer
github.com/fmagno · linkedin · stackoverflow
flmagnom@gmail.com · Portugal · Remote

Summary

Generalist software engineer and consultant with over a decade of experience designing, building and running cloud-native systems, with a preference for Python and AWS serverless but not limited to them. Works from the client's problem: gathers requirements, designs the architecture and records the decisions. Designs microservices to stay small, loosely coupled and independently deployable, and avoids the distributed monolith. Puts deliberate effort into both the development workflow (test-driven, peer- and AI-reviewed) and the deployment workflow (CI/CD), so quality is provable before anything reaches production. Observability is a first-class citizen: logs, metrics and traces break each business flow into steps that can be followed and scrutinised, and dashboards keep services and flows visible and under control. Above all, communicates clearly with stakeholders before, during and after delivery. Experienced with multi-tenant platforms and data residency constraints, with deployments across Europe, North America and Australia. AI is present at every stage of the engineering process. Comfortable owning services end to end, from design to production. Has lived and worked in the UK, Portugal and Angola, often in multicultural teams. Fluent in English and Portuguese.

AWS · Serverless · CDK | Python · JavaScript · AsyncIO | GitHub Actions · CircleCI | Datadog · Dynatrace | TDD · Microservices | Agentic AI · T3 Code · herdr

Experience

TELUS Health

Nov 2023 – Present · Remote

Senior Software Engineer (contract)

Backend engineer in Internal Product Analytics, the team that owns the platform's event pipelines and the reporting data lake behind BI and product decisions. Its mission is to drive and enable data-driven decision making.

Assessment and Appointment flows tracking · Click-event ingestion for an enterprise client launch

SOLE BACKEND ENG

Captures click-level events from the assessment and appointment-booking flows of the mental-health web app, for an enterprise client, replacing Google Analytics and Datadog RUM, which could not show per-step behaviour.

IMPACT Design and implementation of the whole backend from scratch, working with product, front-end, identity, DevOps and QA teams against a challenging launch date.

STACK Python · AWS CDK · API Gateway (Lambda authoriser, VTL to SQS) · SQS · Lambda · DynamoDB · Firehose · S3 (Parquet) · Glue · Athena · GitHub Actions (OIDC) · Dynatrace

Global Intake Form · Global Intake Form ingestion · GraphQL API

SOLE BACKEND ENG

Backend of the Global Intake Form in TELUS Health One, which lets employee-assistance members in countries without online scheduling request a confidential counselling appointment, in up to 60 languages. Each request is stored and routed to the Global Services Team.

IMPACT Design and implementation of the backend for the initial build and launch, working with product, QA and front-end teams. Live since 1 July 2025 in the EU, AU, CA and US, serving hundreds of requests in the first month.

STACK Python · FastAPI · Strawberry GraphQL · Pydantic · Lambda · API Gateway (IAM) · DynamoDB · EventBridge · Secrets Manager · CloudWatch · X-Ray · Apollo Federation · Serverless Framework · GitHub Actions · Datadog

Reporting pipeline · Event ingestion for BI

PRIMARY DEVELOPER

Captures business events from various producers and channels them into a data lake consumed by the business-intelligence team. Main operations: validation, PII redaction, transformation and storage.

IMPACT Expanded support to additional event types and modernised the reporting pipeline by replacing home-grown logic and services with specialised tools and native components: Pydantic, FastAPI, AWS EventBridge, Datadog and, later, Dynatrace.

STACK Python · Pydantic · Lambda · EventBridge · SQS · SNS · Firehose · S3 (Parquet) · Glue · Athena · Terraform · Serverless Framework · GitHub Actions · CircleCI

Cross-cloud event routing · AWS ⇄ Azure

PRIMARY DEVELOPER

Routing pipeline that bridges business events between AWS and Azure: appointment, consent, organisation, payment and session events flow in both directions between AWS EventBridge and the business systems that run on Azure, across four regions.

IMPACT Expanded support to additional event types and modernised the routing pipeline by replacing home-grown logic with specialised tools and native components: Pydantic, W3C trace context, AWS EventBridge, Datadog and, later, Dynatrace. Moved outbound delivery off Solace to an Azure RDP endpoint.

STACK Python · Pydantic · Lambda (arm64) · EventBridge · CloudEvents · Typer · Serverless Framework · GitHub Actions · Datadog · Dynatrace

Dynatrace migration · Observability · Datadog to Dynatrace

MIGRATION OWNER

Responsible for migrating all of the Internal Product Analytics team's services from Datadog to Dynatrace, as part of the company-wide consolidation of observability: traces, logs, metrics, dashboards and alerts.

IMPACT Migrating the services to Dynatrace on schedule.

STACK Dynatrace

Earlier experience

2009 – 2023

2021 – 2023 **Coquet Consulting** Backend Engineer (contract) — Remote, Portugal
Backend contracts for several clients in async Python (FastAPI, SQLAlchemy 2.0, PostgreSQL) and NestJS on GKE. Integrated banking, HR, shipping and e-commerce APIs (Solaris, Workday, Personio, FedEx, Shippo, Shopify, SendGrid), including OAuth2 flows.

2017 – 2018 **Unity Technologies** Software Engineer — UK
Provisioning and test services for the Unity engine: Go and Python over gRPC and NSQ on OpenStack, with Raspberry Pi hardware-in-the-loop rigs.

2012 – 2016 **Imagination Technologies** Software Engineer — UK
Async task scheduler and device allocator for GPU hardware validation; PowerVR Rogue driver test framework; virtualised CI on Xen and KVM/QEMU. Led test planning with a team in India. Python, Django, Puppet.

2011 – 2012 **Bullray** Research Engineer — Angola
Gyro-stabilised gimbal and off-grid traffic and coastal surveillance systems. LabVIEW, C/C++, SolidWorks.

2009 – 2011 **Bosch Security Systems** QA Engineer — Portugal
Test jigs for surveillance and communication systems (LabVIEW, Eagle PCB). Before that, researcher at Instituto de Telecomunicações on predistortion models for RF power amplifiers (MATLAB).

Independent work

github.com/fmagno

Agent workspace (Claude Code, Pi, MCP, TypeScript, 2026) · chatbot-fm (LlamaIndex, ChromaDB, Ollama, 2026) · surl (FastAPI, SQLAlchemy 2.0, Postgres, 2023) · nbapp (FastAPI, Next.js, MinIO, tusd, 2022)

Core skills

LANGUAGES	Python (asyncio, Pydantic, FastAPI, Strawberry, pytest), TypeScript / NestJS, Go, SQL.
AWS & EVENT DESIGN	Lambda, API Gateway, SQS/SNS, EventBridge, Firehose, DynamoDB, Glue, Athena, CDK, Terraform, data residency.
DELIVERY & SECURITY	GitHub Actions, CodeQL, Wiz, tag-based releases.
OBSERVABILITY	Datadog and Dynatrace for logs, metrics, traces and dashboards; trace-context propagation across services and clouds; structured, PII-safe logging.
AI-ASSISTED ENGINEERING	Claude Code and Pi agents with custom skills, MCP servers and hooks; local RAG with LlamaIndex.

Education & languages

University of Aveiro · MSc Electronics & Telecommunications Engineering

Portuguese (native) · English (fluent)